

Solution for Chapter 4

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4.1

$$\begin{aligned} E_{S \sim D^m}[\hat{R}_S(h_S^{ERM})] &= E_{S \sim D^m}[\inf_{h \in \mathcal{H}} \hat{R}_S(h)] \\ &\leq E_{S \sim D^m}[\hat{R}_S(h)], h \in \mathcal{H} \\ &= R(h), h \in \mathcal{H} \\ &\leq \inf_{h \in \mathcal{H}} R(h) \end{aligned}$$

$$\begin{aligned} E_{S \sim D^m}[R(h_S^{ERM})] &\geq E_{S \sim D^m}[\inf_{h \in \mathcal{H}} R(h)] \\ &= \inf_{h \in \mathcal{H}} R(h) \end{aligned}$$

4.2

$$\begin{aligned} \Phi(u_1) + \Phi(u_2) &= (1 + u_1)^2 + (1 + u_2)^2 \\ &= u_1^2 + u_2^2 + 2(u_1 + u_2) + 2 \\ &\geq \frac{1}{2}(u_1 + u_2)^2 + 2(u_1 + u_2) + 2 \\ &= 2(1 + \frac{u_1 + u_2}{2})^2 \\ &= 2\Phi(\frac{u_1 + u_2}{2}) \end{aligned}$$

So, $\Phi(u)$ is convex and non-decreasing when $u \geq -1$. Put $s = 2$ and $c = \frac{1}{2}$ into theorem 4.7 and get the objective inequality.

4.3

$\Phi(u)$ is convex and non-decreasing when $u \geq -1$. Put $s = 2$ and $c = \frac{1}{2}$ into theorem 4.7 and get the objective inequality.

4.4

(a) $\eta(x) = P[y = +1|x]$

Bayes classifier is a function that get +1 when $\eta(x) \geq \frac{1}{2}$, -1 when $\eta(x) < \frac{1}{2}$.

Bayes scoring function $h^*(x) = \eta(x) - \frac{1}{2}$.

(b)

$$R(h) - R^* = 2E_{x \sim D_x} [|h^*(x)| \mathbb{1}_{h(x)h^*(x) \leq 0}]$$